atomQUEST

Round 3

Proto Submission: Level 2 Submission

Instructions

Submission Requirements:

1. Video Prototype:

a. 90-second video showcasing:

- Final team members (only if changed from Round 2) (if team members are in different locations, mention this clearly in the video) (Refer to Annexure I)
- Your bot in motion
- Your bot cleaning dust /garbage as defined in the problem statement
- Any additional features that are differentiators and have a relevant use case

b. The duration of the video should **not exceeding 120 sec**

c. Be uploaded to a Google Drive folder along with related pictures (final bot, components, sketches, or any supporting visuals). Along with the video, the following is **also mandatory:**

1. One pager specification sheet of your bot (example specs: battery life/ operation duration in single charge, motor power, suction capacity, dimensions, control type – autonomous or manual, charging time, battery capacity, etc.)
2. One page block diagram showing the major subsystems in your vacuum cleaner

d Make sure to keep the drive link open for all, otherwise we won't be able to access , hence will not consider your submission for evaluation.

2. **Mentor's Review & Feedback**
Each team must submit a signed review note from their faculty mentor / professional. The note should include the **date of review, mentor's observations and feedback, and their signature**. This review will NOT be a deciding factor in evaluation but will help us understand the mentor's involvement, whether the work was reviewed, and how their inputs supported the team in refining and improving their project (refer to Annexure II). A thorough review process yields a sharper output hence this is required

3. Google Drive Folder:

- Must contain Video of the bot cleaning the dust /garbage as defined in the problem statement
- Mentor's review and feedback form
- Images/documents supporting feasibility (Bot photos, component invoices, design sketches, etc.)

4.Submission Details:

1.The google drive link needs to be shared as final deliverable.
2.Make sure to keep the drive link open for all, otherwise we won't be able to access , hence will not consider your submission for evaluation.
3. Follow the naming convention strictly for your report and video submissions.

Naming Convention for Mentor Review and Feedback Form :

AtomQuest2025_[TeamName]_[CollegeName]_MentorConsent

Example: AtomQuest2025_TechInnovators_IITDelhi_MentorConsent

Naming Convention for Video:

AtomQuest2025_[TeamName]_[CollegeName]_Video

Example: AtomQuest2025_TechInnovators_IITDelhi_Video

Naming Convention for Folder with Images :

AtomQuest2025_[TeamName]_[CollegeName]_Images

Example: AtomQuest2025_TechInnovators_IITDelhi_Images

Naming Convention for Drive :

AtomQuest2025_[TeamName]_[CollegeName]_Folder

Example: AtomQuest2025_TechInnovators_IITDelhi_Folder

Submissions that do not adhere to the format will not be considered for evaluation.

5.Deadline:

- The project report and video submission deadline is **15th October.**
- **Advisory:** It is strongly recommended that you finalize your bot by **10/12th October** to allow sufficient time for submission and your internal review .

Further Round Details

Round 4: Grand Finale

- The **top 10 teams** will be shortlisted to present their solutions at Atomberg's R&D centre to R&D leadership
- **Date:** Final dates to be announced.
- **Venue:** Atomberg Innovation Center, Pune

- **What will be known to you?**

    1. Shape and dimensions of the room
    2. Nature and number of obstacles
    3. Types of garbage

- **What will be unknown?**
    1. Arrangement of obstacles in the room
    2. Arrangement of garbage in the room

- **Competition Flow**
    1. Between two teams, the challenge room will be reset by the ATPL team
    2. No two teams will get the same arrangement of obstacles and garbage.
    3. The team will be given 1 minute to inspect the room before placing their bot at the docking station
    4. Once the round starts, the bot needs to start from the docking station, and go about cleaning the room. Once it is done cleaning, it should return to the docking station. Failing to return to the docking station will mark your attempt as incomplete.
    5. Each team will be given a score (out of 100) based on the following factors (the weightage for each will be informed beforehand)
        a. Completion time (20%)
        b. Efficiency for each garbage type (w/w cleaning ratio) (25%)
        c. Whether autonomous or manual (wireless controlled) (20%)
            i. Obstacle avoidance effectiveness
        d. Battery life (15%)
        e. Aesthetics (5%)
        f. Ease of removing and reattaching the storage container (10%)
        g. Cost of the system (5%)

- **Constraints**
    1. The total time allotted for the cleaning session will be 5 minutes.
    2. If your bot gets stuck in-between, you can take it back to the docking station and restart the challenge. The timer keeps going forward; the first restart will reduce your challenge time by 5 seconds, the second by 10 seconds, and so on.
    3. The bot dimension should be within these bounds:
        a. Length: 1.5 ft
        b. Breadth: 1.5 ft
        c. Height: 1 ft

(*Further details will be shared closure to the round)*

Contact: For any queries, email us at atomquest2024@atomberg.com

Annexure I– Sample Video Structure

[Intro – 15 sec – Only if team has changed]

- State your **Team Name**.
- Introduce all final team members (Name, College, Department).
- If members are in different locations, mention that clearly.

[Idea Overview – 30 sec]

- Mention the **Top 3 specifications** of your bot that make it unique.
- Highlight what differentiates your bot from others.

[Bot in Action – 30 sec]

- Show your bot performing **motion and cleaning** (mandatory).
- Go beyond the basics—highlight **at least one unique aspect** (e.g., efficiency, adaptability, design innovation, smart control, etc.).

[Path to Finale – 10 sec]

- Share your **timeline and milestones** leading to the finale.
- If any **major change or upgrade** is planned (e.g., converting from remote-controlled to autonomous), explain how you will achieve it.

[Closing – 5 sec]

- Wrap up with a short thank you / sign-off.

Important Note for Teams

*This round will have a **higher elimination ratio**. Use this opportunity to make a **strong case for why your bot deserves to be in the Finale**. While motion and cleaning are basic requirements, we strongly encourage you to showcase **unique features and innovations** that set your bot apart.*

Annexure II – Mentor Review & Feedback Form

Team Name: _____
Mentor Name & Designation: _____
Date of Review: _____

Assessment

A. Strengths / What is working well
(Briefly describe key positives observed in the team's work)

B. Areas of Improvement
(Briefly describe gaps, challenges, or suggestions for enhancement)

C. Mentor's Inputs / Guidance Provided
(Briefly outline how you guided the team and the kind of support shared)

Declaration:
This is to confirm that I have reviewed the team's work. In my assessment, I have noted what aspects are working well, where improvements are needed, and how my inputs supported the team in refining their approach.

Signature of Mentor: _____

Annexure III – Faculty Mentor Consent Letter

(To be printed on Institute Letterhead)

Date: _____

To,
The Organizing Committee
Atomberg Technologies - Atomquest 2025

Subject: Faculty Mentor Consent and Support Confirmation

Dear Organizing Committee,

This is to confirm that I, **Prof./Dr. _____**, faculty member of **(Department/School/Institute Name)**, am the official Faculty Mentor for the team **_____** participating in **Atomquest 2025** organised by Atomberg Technologies .

I hereby extend my full consent and support to guide and mentor the team throughout the course of the competition, starting from the registration phase until the **Finale** of Atomquest 2025. I undertake to be available for necessary academic guidance and extend support to ensure the team's active participation in the competition.

I also acknowledge that I am aware of the rules, timelines, and expectations of Atomquest 2025, and I confirm my commitment as Faculty Mentor for the aforementioned team.

Sincerely,

Name of Faculty Mentor: _____
Designation: _____
Department/School/Institute: _____
Contact Number: _____
Email ID: _____

Signature with Seal/Stamp